BEER GAME 2.0



PLAY GAME

ABOUT

QUIT

Login to Play

Enter your credentials to access the game

Username

Enter your username

Password

••••••••

Login

Game Configuration

Configuration Summary

Players: 1 Human, 3 AI
AI: Base-Stock
Scenario: Step Change
Duration: 25 weeks

Start Game

1. Number of Players

1 2 3 4

2. Demand Scenario

Step Change
Demand: 4 → 8 units at week 5

Random (Poisson)
Variable demand, λ = 6 per week

Trend + Noise
Growing demand: 4 + 0.4t + noise

3. Assign Roles

Retailer	Human
Wholesaler	AI
Distributor	AI
Factory	AI

4. Advanced Options

Week 7 of 25

Retailer

Human

This Week – Your State & Order

Current Status

Customer demand this week: 8 units

Inventory at start of week: 12 units

Backlog at start of week: 2 units

Order to upstream partner

− 10 +

ORDER!

After this week (results)

Inventory at end of week: 14 units

Backlog at end of week: 0 units

Holding cost this week: 14 €

Backlog cost this week: 0 €

Total cost this week: 14 €

Cumulative: 98 €

Advance Week →

Shipments – Outgoing and Incoming

You ship this week

Shipment to partner: 8 units

Leaves inventory now; arrives week 8

Incoming to you

Arriving next week (t+1): 10 units

Arriving in 2 weeks (t+2): 12 units

Timeline

Week 7 — You are here

Order: 10 units

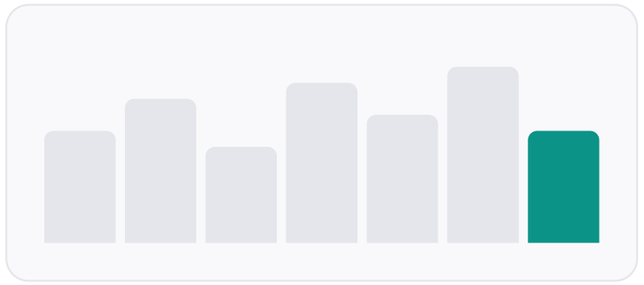
Ship: 8 units

Week 8

+10 units arriving

Progress 7 / 25 weeks

History & Totals



Total holding cost: 220 €

Total backlog cost: 480 €

Total cost so far: 700 €

Start of Next Week

Expected Inventory

16

units

= 14 units at end of this week + 10 units arriving next week

Exit Game

Game Complete!

Week 25 of 25 • Scenario: Step Change

Overall Results

Total Cost

💰 2,450 €

Average Inventory

📦 15.3 units

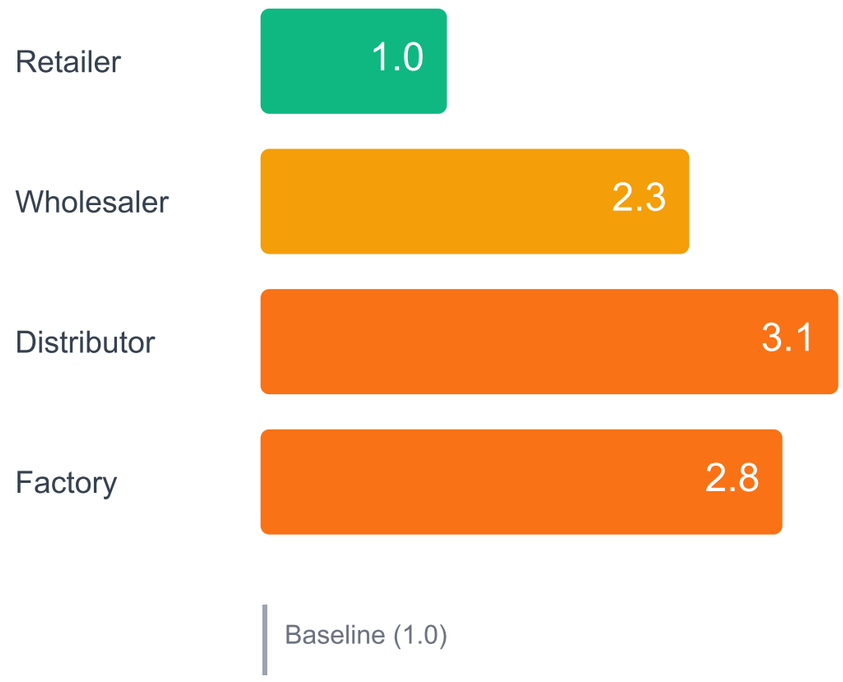
Service Level

78%

★★★☆☆

Good performance

Bullwhip Effect Analysis



Per-Tier Breakdown

Tier	Total Cost	Avg Inventory	Avg Backlog	Bullwhip Ratio
Retailer	520 €	10.2	2.1	1.0
Wholesaler	680	14.5	3.8	2.3
Distributor	750 €	18.1	4.2	3.1
Factory	500 €	16.8	2.5	2.8

Key Insights

- Bullwhip effect detected: 3.1× amplification from Retailer to Distributor
- Wholesaler experienced highest average backlog (3.8 units)
- Recommendation: Try PI Controller policy to reduce oscillations
- Factory maintained stable costs through consistent production

View Analysis | Play Again | Export CSV | Exit